

Candidate Seat No: _____

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Master of Computer Applications (MCA)
Semester-I University Examination March -2018
CA721 – Fundamental Concepts in Programming Language

Date: 28/03/2018, Wednesday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 70

Instructions:

1. Attempt the two sections in separate answer books.
2. Figure to the right indicate marks.
3. Make suitable assumptions wherever necessary and state them.

SECTION-I

Q.1 Do as directed :

1. Define Variable and Datatype. [2]
2. List any 4 Reserve keyword available in C. [2]
3. “C Language” was developed by _____. [1]
4. Execution of the C Program begins with _____. [1]
5. Write syntax of FOR Loop. [2]
6. Define Pointer. [1]
7. sizeof(int) returns how many bytes? [1]
8. Write syntax of IF...ELSE statement. [1]

- Q.2[A]** Write a program to determine whether the given number is positive or negative. [3]
[B] Write Sample C program with explanation. [5]
[C] Describe data type available in C. [4]

OR

- Q.2[A]** Write a program to swap two numbers using temporary variable. [3]
[B] List out types of operator available in C. Explain any two with example. [5]
[C] Write note on structure of C program. [4]

- Q.3[A]** How we can declare and initialize variable in C? Also discuss rules for declaring variable. [3]
[B] Write a notes on IF...ELSE..IF statement. [4]
[C] What is an Escape Sequence? What is its purpose? Explain any two Escape sequence with example. [5]

OR

- Q.3[A]** Differentiate between Structure and Union. [3]
[B] Write a program to search element from the array. [4]
[C] Write a program to find minimum and maximum element from the array. [5]

SECTION-II

Q.4[A] Do as directed:

1. Give example of Pointer declaration and initialization. [2]
2. Write syntax of **SWITCH** Statement. [2]
3. The type of all elements in array must be same. [True/False] [1]
4. What is extension of C Program file? [1]
5. Char data type in C Language occupy _____ byte in memory. [1]
6. Differentiate %c and %s format specifier. [1]

[B] Differentiate between Entry Controlled and Exit Controlled loop. [3]

Q.5[A] What is Stack? Explain algorithm of all operations of stack. [8]

[B] List out types of User Define Function. Explain any one with example. [4]

OR

Q.5[A] What is Queue? Explain algorithm of all operations of queue. [8]

[B] List all String functions available in C and explain any one in detail. [4]

Q.6 Answer the following.(Any three) [12]

[A] Write a C program to display following pattern.

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

[B] Differentiate between NULL Pointer and Generic Pointer.

[C] Create a structure **Student**. Store 5 student's information and display information in proper format.

[D] Read a String and display length of the string.

[E] Write a note on Array.